

Research paper

Numerical investigation of combustion characteristics of hydrogen-enriched low calorific value landfill gas for energy applications

Udaya Kahangamage^{a,*}, Kangdong Chen^a, Haisheng Zhen^b, Chun-wah Leung^a

^a School of Professional Education and Executive Development, The Hong Kong Polytechnic University, Hong Kong

^b The Mechanical and Electrical Engineering College, Hainan University, Haikou, China

ARTICLE INFO

Keywords:

Low calorific value landfill gas
Hydrogen-enriched fuel
Laminar burning velocity
Alternative energy

ABSTRACT

Landfill gas is regarded as a promising renewable energy resource. However, Low calorific value landfill gases with less than 40 vol% CH₄ are not well suited for energy applications and therefore disposed by flaring or direct emission to atmosphere causing environmental problems. In the present work, the potential of using hydrogen-enriched low calorific value landfill gas for energy applications has been explored through numerical investigation of combustion characteristics of different blends of fuels. The ignition delay of fuel mixtures decreased with the increase of hydrogen fraction (H₂ enrichment from 0 to 50 vol%) at stoichiometric conditions. At high initial pressure conditions, ignition delay showed a weak relationship with the pressure. The laminar burning velocity increased with the increment of hydrogen fraction for the range of equivalence ratios considered ($\varphi = 0.7 - 1.4$) and also with the increase of initial temperature. However, the laminar burning velocity decreased significantly with the increasing initial pressure and hydrogen enrichment has insignificant effect at high initial pressures. The sensitivity analysis shows that the main elementary reactions inhibit the laminar burning velocity more with increasing initial pressure. The combustion performance of low calorific value landfill gas with 30 vol % CH₄ enriched with 30 vol% hydrogen shows comparable performance to biogas with 60 vol% CH₄ for a range of operating conditions. The numerical investigation results show that hydrogen-enriched low calorific value landfill gas could be used for thermal applications with the use of suitably designed combustion systems operating at low initial pressures and relatively high initial temperatures.

1. Introduction

Energy systems used for energy needs are rapidly evolving to reduce environmental impacts and face the challenges related to worsening climate change, pollution, fast depleting non-renewable energy sources and the growing global population. In the quest for expanding the sources of energy, various alternative fuels with low calorific value are gaining more attention (Stančin et al., 2020; Kumar et al., 2020; Askari and Ashjaee, 2017; An et al., 2023). Methane containing landfill gas (LFG) is one of those promising alternative fuels. LFG has an advantage over other renewable fuels like syngas as it is derived from waste with minimal processing. The composition of the LFG varies widely depending on the design, age, and the deposited content of the landfill sites. LFG generated from a mature site typically consists of 50–65 vol% of methane (CH₄), 30–50 vol% of carbon dioxide (CO₂), 2–5 vol% of nitrogen (N₂) and some traces of other substances. The higher heating value of pure methane is 39.82 MJ/m³. However, the heating value of

LFG containing 60 % CH₄ is 22.6 MJ/m³ (refer to Fig. 1) due to the presence of non-combustible gases such as CO₂ and N₂. Currently, there are well-developed technologies to make use of LFG with higher calorific value for power generation and to produce synthetic natural gas that can be fed into existing gas supply networks. However, LFG with low to medium calorific values are still disposed by flaring or direct emission to atmosphere. The most common sources of low calorific value LFG are older closed landfills, shallow waste deposits subject to ambient conditions, perimeter gas collection systems, etc (David Browell et al., 2009).

With the use of appropriately designed burners and employing fuel enhancement techniques, low calorific value gaseous fuels, such as LFG can be used for various energy applications. One of the common uses of LFG is onsite evaporation of leachate to a more concentrated and more easily discarded form (U. S. E. P. A. EPA, 2024). It can also be used as a supplementary fuel for industrial boilers and as a reburning fuel to reduce emissions of pollutants from combustion processes (Uchman and Werle, 2016). The major challenge with the low calorific value LFG is the presence of a higher percentage of inert gases (diluent) that inhibit

* Corresponding author.

E-mail address: udaya.kahangamage@cpce-polyu.edu.hk (U. Kahangamage).

<https://doi.org/10.1016/j.egy.2024.06.008>

Received 13 January 2024; Received in revised form 2 April 2024; Accepted 4 June 2024

Available online 15 June 2024

2352-4847/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Nomenclature

ρ	gas density, kg/m ³
C_p	specific heat at constant pressure, J/(kg·K)
u_x	flow velocity in x direction, m/s
LBV	laminar burning velocity, cm/s
θ	angle between axis of flame and flame front, rad
T	temperature, K
P	pressure, Pa
λ	thermal conductivity, W/(m·K)
φ	Equivalence ratio
τ	non-dimensional temperature
τ_i	ignition delay time
H	flame height, mm
r	radial distance from axis of flame, mm
z	axial distance from axis of flame, mm
R_c	the mole fraction of carbon dioxide in Biogas, $n_{CO_2} / (n_{CH_4} + n_{CO_2})$
R_H	the mole fraction of hydrogen in fuel, $n_{H_2} / (n_{CH_4} + n_{CO_2} + n_{H_2})$
S_{LBV}	normalized sensitivity coefficient

narrowed range of flame stability and lowered combustion efficiency. The potential of enhancing combustion performance of biogas by enriching it with different percentages of other fuels such as propane, butane, LPG, natural gas, and hydrogen are also reported (Cardona and Amell, 2013; Chandra et al., 2011; Kahangamage et al., 2022, 2021).

Hydrogen enrichment of biogas has been proven to enhance combustion performance while reducing adverse environmental impacts (Li et al., 2017; Wei et al., 2019; Yuyin et al., 2009; Hu and Zhang, 2019; Wei et al., 2020, 2016; Chen et al., 2021). Hydrogen is a zero-carbon fuel with calorific value of 120 – 142 MJ/kg (10.8 – 12.7 MJ/m³) and a wide range of flammability (4 % - 75.6 %) (David Browell et al., 2009). Its adiabatic flame temperature is 2127 °C and only combustion product is water vapor (H₂O). Li et al. (2017) conducted a numerical investigation of combustion and heat release characteristics of biogas non-premixed flames with various fractions of hydrogen enrichment using the GRI-Mech 3.0 mechanism and P1 radiation model. The reported results showed that the net heat release rate of biogas flame increased with the increasing hydrogen addition ratio. Wei et al. (2019) investigated the laminar burning velocity of biogas-hydrogen mixtures with methane content 40 – 60 vol% for various equivalence ratios using cylindrical combustion bomb apparatus. They have also performed numerical investigations with GRI-Mech 3.0 and San Diego mechanisms. The laminar burning velocities of all the biogas samples tested showed almost a linear increase with the addition of hydrogen up to the equivalence ratio

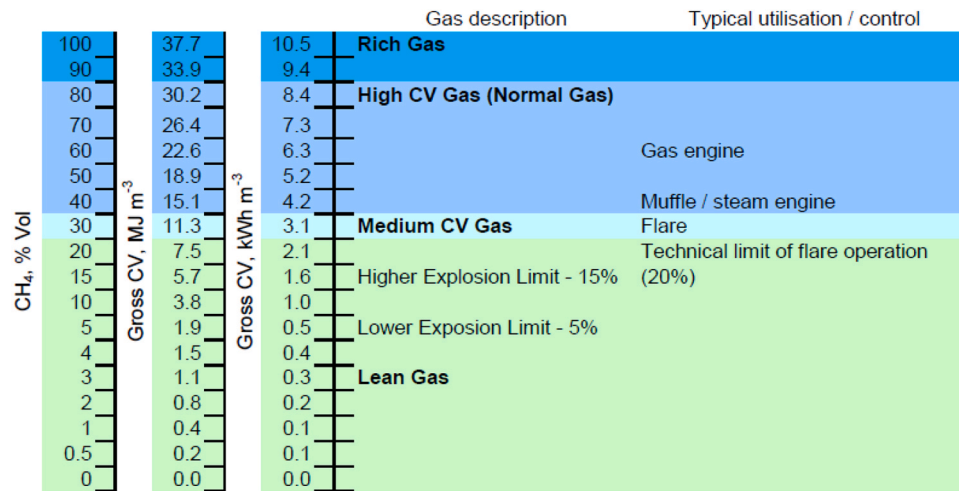


Fig. 1. Methane content and calorific values for landfill gas (David Browell et al., 2009).

efficient combustion through various chemical and physical interaction mechanisms.

The wide variation of the constituents in the LFG causes a considerable variation in combustion characteristics and hence presents many challenges in its use (Askari and Ashjaee, 2017). Therefore, it is essential to have accurate estimations of fundamental combustion characteristics of LFG with wide variation of constituents if we plan to use them for thermal applications. There are many research studies to study the fundamental combustion characteristics of biogas with high calorific values (Hinton and Stone, 2014; Dai et al., 2012; Zhang et al., 2016). The main focus of those studies is to investigate the important fundamental characteristics such as laminar burning velocity, ignition delay, flame temperature, active species concentration, heat release rate and flammability limits using experimental and numerical approaches. In addition, the negative effects of CO₂ on combustion stability have been reported extensively in previous studies (Khan et al., 2017; Duva et al., 2020). These previous studies found that CO₂ can effectively radiate and reabsorb thermal energy, and substantially reduce the laminar burning velocity, resulting in reduced flame temperature and burning rates, a

of 1.1. There is a downward trend for rich mixtures. The experimental results matched more with the predictions provided by GRI-Mech 3.0 mechanism.

There are few research studies specially focusing on gaining better understanding of combustion performance of different LFG compositions. Hinton and Stone (2014) investigated the LBV of high calorific value LFG containing 40 vol% CO₂ and 60 vol% CH₄ using constant volume combustion bomb with the Schlieren system. This investigation covers the pressure range of 1 – 4 bar and equivalence ratio range of 0.7 – 1.4. Z Zhang et al. (2016). conducted a comparative study to analyse the effects of CO₂ and N₂ dilution on methane/air flames. Their investigation was limited to the CO₂ and N₂ concentrations up to 20 vol% and concluded that the dilution shows a weak effect while the change in concentrations of reactive species formed in the flame play an important influence on the progress rate of combustion reactions. Qin et al. (2001). conducted a comprehensive investigation on fundamental combustion parameters and environmental aspects of LFG as applied to power generation with high calorific value gas mixtures. The results show that the presence of CO₂ in LFG significantly decreases the LBV, extinction

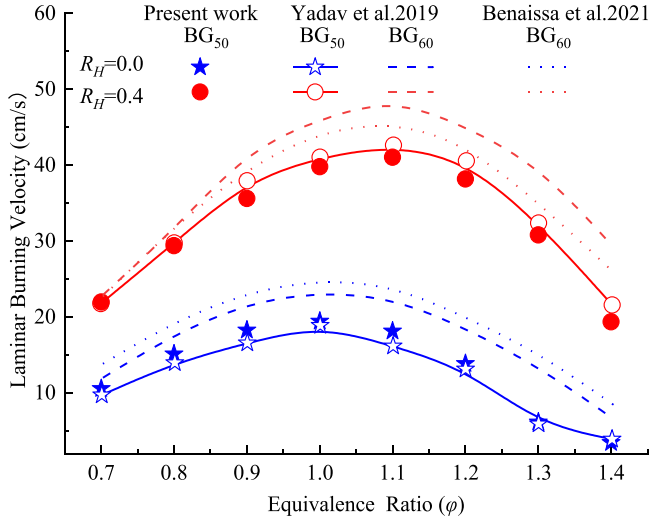


Fig. 4. Comparison of predicted laminar burning velocity of biogas-air and biogas/hydrogen-air mixtures and reference values from literature (Benaissa et al., 2021; Kumar Yadav et al., 2019a) at ambient conditions ($T = 298$ K and $P = 1$ bar).

$$\rho u \frac{dY_k}{dx} + \frac{d}{dx}(\rho Y_k V_k) - \dot{\omega}_k M_k = 0, k = 1, 2 \dots K \quad (4)$$

Where, M_k is mole mass of species k .

3. Simulation validation

3.1. Ignition delay

It's meaningful and necessary to investigate the ignition delay of fuel-oxidizer mixtures, because optimal ignition delay is key to avoid pre-ignition and delayed ignition in combustion systems. In a previous study, Zeng et al. (2015) reported experimentally measured ignition delay times of methane/air mixtures diluted with CO_2 using chemical shock tube method. Part of this experimental data is used to validate the simulation model developed in the present study. As reported in reference Zeng et al. (2015), there exists a linear relationship between the reciprocal of temperature and the logarithm of the ignition delay times. Fig. 3 shows the comparison between the experimental data and the simulation data of present study using GRI-Mech 3.0 reaction mechanism over the temperature range of 1300–1800 K, at pressure of 1Mpa and equivalence ratio $\phi=0.5$ for the CO_2 dilution coefficients of 20 %

and 50 %.

As shown in Fig. 3, prediction of ignition delay times of simulation is in better agreement with that of experimental data. Similar accurate prediction is also reported at equivalence ratio $\phi=1$ and initial pressure of 0.1 MPa by Benaissa et al. (2021). It shows the effectiveness of numerical model used and GRI-Mech 3.0 mechanism in capturing the combustion characteristics of CH_4 -air mixtures diluted with CO_2 such as LFG-air mixtures at different operating conditions.

3.2. Laminar burning velocity

Fig. 4 shows the experimental (Kumar Yadav et al., 2019a) and computational Laminar Burning Velocity (LBV) of biogas and H_2 -enriched biogas flames at 1 bar and 298 K. The opened symbols and closed symbols-line in Fig. 4 represent the simulated and experimental data for LBV, respectively. The LBV of biogas (50 % CH_4 -50 % CO_2 , hereinafter referred to as BG_{50}) with 0 % and 40 % H_2 enrichment ($R_H=0.0$ and $R_H=0.4$) were studied computationally by using Ansys Chemkin-Pro with GRI-Mech 3.0. The numerical prediction results are in better agreement with experimental data reported by Yadav et al (Kumar Yadav et al., 2019a).

The data in Fig. 4 shows that the LBV of BG_{50} increases with increase in equivalence ratio (ϕ) till $\phi=1$ and then starts decreasing with increasing ϕ . With the H_2 enrichment, maximum LBV occurs when the fuel mixture becomes slightly rich ($\phi=1.1$). As explained by Benaissa et al. (2021), this is due to the changes of average heat release rate (HRR₀) and the average thermal diffusivity (α_0) with the addition of hydrogen. The LBV is proportional to square root of product of HRR₀ and α_0 over the flame thickness. As shown in Fig. 5, with the addition of hydrogen, both HRR₀ and α_0 peaked closer to $\phi=1.1$. Hence, there is a shift in the maximum LBV of hydrogen enriched fuel mixture compared with pure BG_{50} . The data also show that the LBV of hydrogen enriched mixture has increased as compared to pure BG_{50} for the range of equivalence ratios studied. It is due to the enhancement in HRR₀ and α_0 with the addition of hydrogen in the fuel mixture. According to the simulation results, the maximum value of LBV for pure BG_{50} -air mixture at stoichiometric ambient conditions is 19.44 cm/s. With the addition of 40 % H_2 ($R_H=0.4$) at stoichiometric conditions, the LBV of the fuel mixture increases to 39.8 cm/s (≈ 2 times of pure BG_{50} -air mixture). The computed LBV of BG_{50} -air mixture ($R_H=0.0$) and BG_{50} /hydrogen-air mixture ($R_H=0.4$) are comparable with the published data for BG_{50} /hydrogen-air mixtures (Benaissa et al., 2021; Kumar Yadav et al., 2019a) as shown in Fig. 4. The LBV of BG_{50} /hydrogen-air flame for $R_H=0.0$ and $R_H=0.4$ are lower than that of BG_{60} /hydrogen-air flame at ambient conditions. Furthermore, the variations of α_0 , HRR₀ and $\sqrt{\alpha_0 \text{HRR}_0}$ of two flames with increase in equivalence ratio are consistent

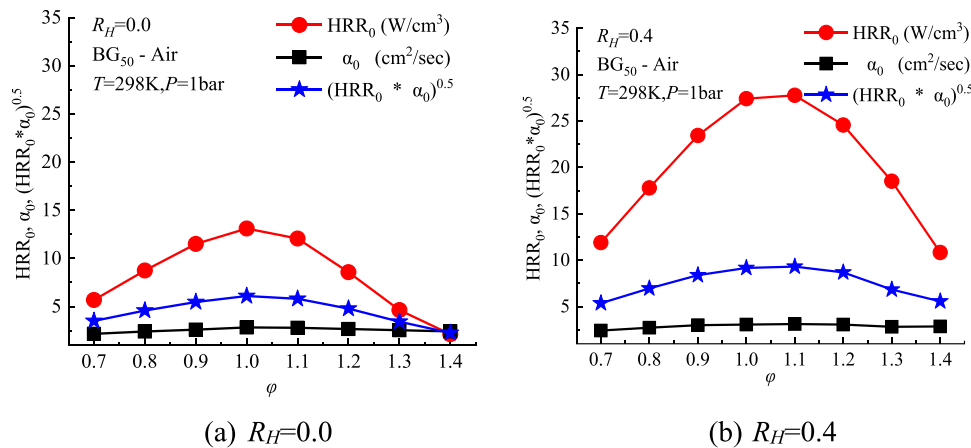
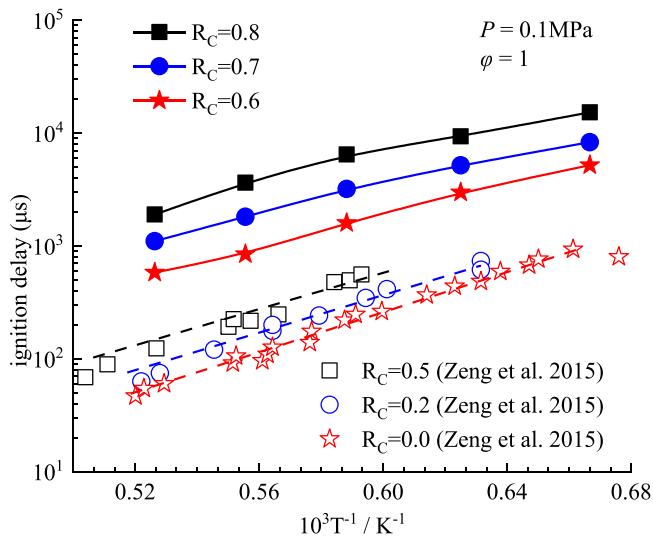
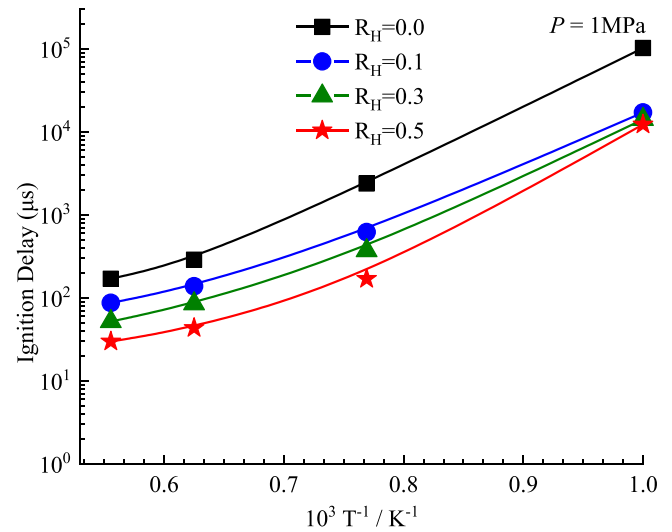


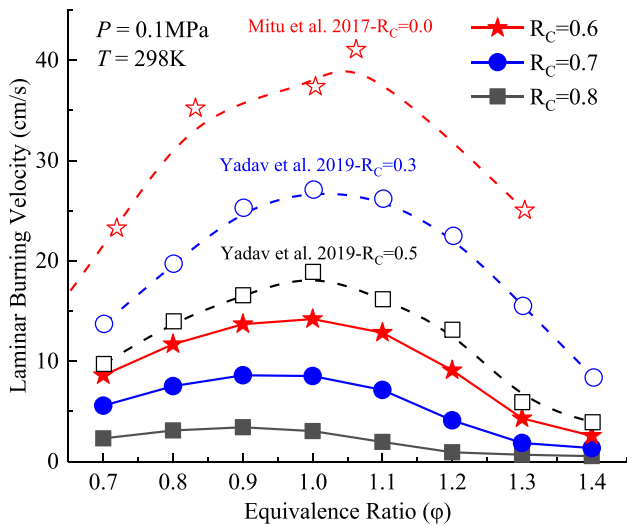
Fig. 5. Variation of average heat release rate (HRR₀), and average thermal diffusivity (α_0) and $(\text{HRR}_0 * \alpha_0)^{0.5}$ with increase in equivalence ratio (ϕ) for two BG_{50} /hydrogen-air mixtures ($R_H=0.0$ and 0.4) at ambient conditions ($P = 1$ bar and $T = 298$ K).



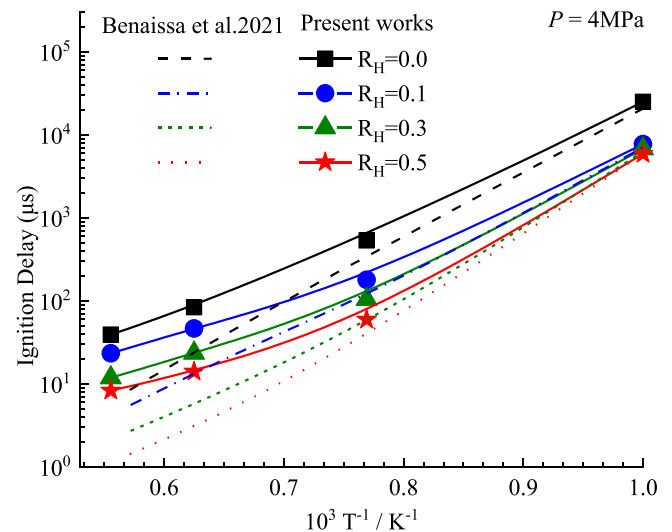
(a) Ignition delay time



(a) P=1MPa



(b) Laminar Burning Velocity



(b) P=4MPa

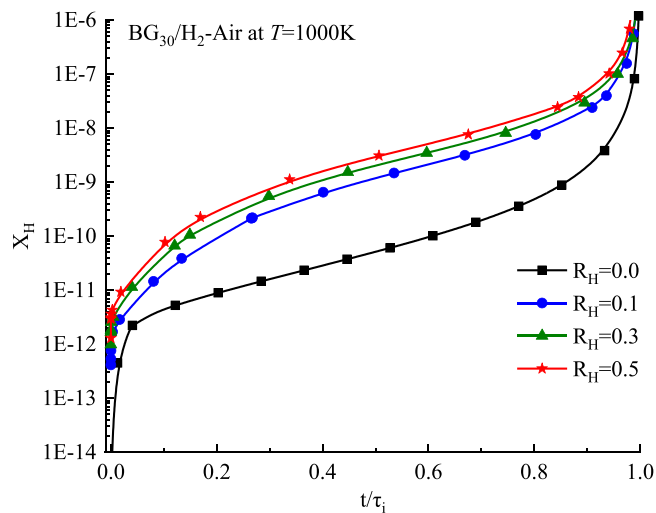
Fig. 6. Variation of Ignition delay time and LBV with an increase in CO_2 fraction added to CH_4 -air mixtures with increase in equivalence ratio (ϕ), at initial temperature 298 K and initial pressure 0.1 MPa. Closed symbols-continuous-lines: computations from current work; Open symbols-dashed-lines: experimental data from literature (Zeng et al., 2015; Kumar Yadav et al., 2019a; Zhen et al., 2019).

as given in Fig. 5. This means that the flame propagation speed of methane is negatively proportional to its dilution with CO_2 . Similar conclusions can also be observed in other references (Benaissa et al., 2021; Kumar Yadav et al., 2019a; Konnov et al., 2018). However, in the literature mentioned above, the proportion of diluted CO_2 in fuel mixture is mostly from 10 % to 50 %, which cannot accurately explain the combustion characteristics of low calorific value methane containing fuel gases with high CO_2 content. The focus of the present study is to investigate combustion performance of hydrogen enriched low calorific value LFG consisting of in excess of 60 % CO_2 by volume. Therefore, it is necessary to study the combustion characteristic of biogas/hydrogen fuel mixtures with fraction of CO_2 in the range of 60–80 % by volume.

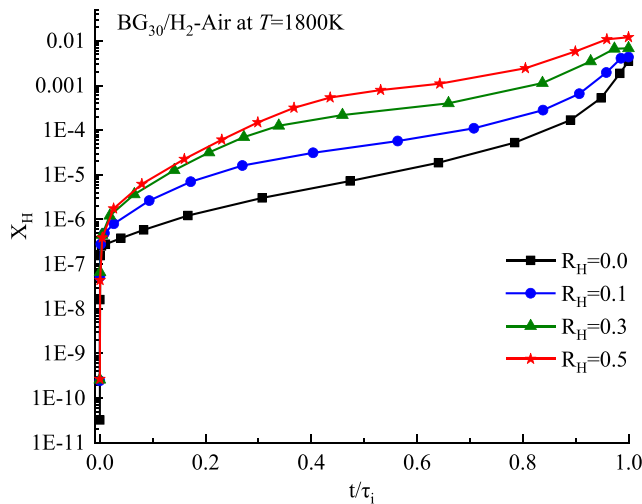
Fig. 7. Variation of ignition delay time with an increase in hydrogen fraction added to BG_{30} -air mixtures at stoichiometric conditions. Closed symbols-continuous lines: computations from present work; Dashed lines: computations for biogas (BG_{60}) from literature (Kumar et al., 2020).

3.3. Effect of increase in CO_2 fraction

There are several experimental and numerical investigations reported in the literature on combustion of CH_4 -air in the presence of diluents such as CO_2 and different inert gases (Benaissa et al., 2021; Zhou et al., 2020; Liu et al., 2020; Wang et al., 2018; Rajesh and Prathap, 2023; Kumar Yadav et al., 2019b; Mitu et al., 2017). Fig. 6(a) shows the comparison of ignition delay data obtained from computations from the current work and the experimental data from the literature. In this paper, dilution of CO_2 is defined as the amount of additional CO_2 in the methane given by $R_c = n_{\text{CO}_2}/(n_{\text{CH}_4} + n_{\text{CO}_2})$. The LBV in various $\text{CH}_4/\text{CO}_2/\text{air}$ mixtures at initial pressure of 0.1 MPa and temperature of 298 K are plotted against the mixture equivalence ratio (ϕ), as shown in Fig. 6(b). The data in Fig. 6(b) show that the LBV of all mixtures increases with increase in ϕ , reach to a peak, and then starts decreasing. The peak value of LBV of pure CH_4 ($R_c=0.0$) occurs at ϕ slightly higher than 1.0 (for a rich mixture). As R_c increases (diluted more with CO_2),



(a) T=1000K



(b) T=1800K

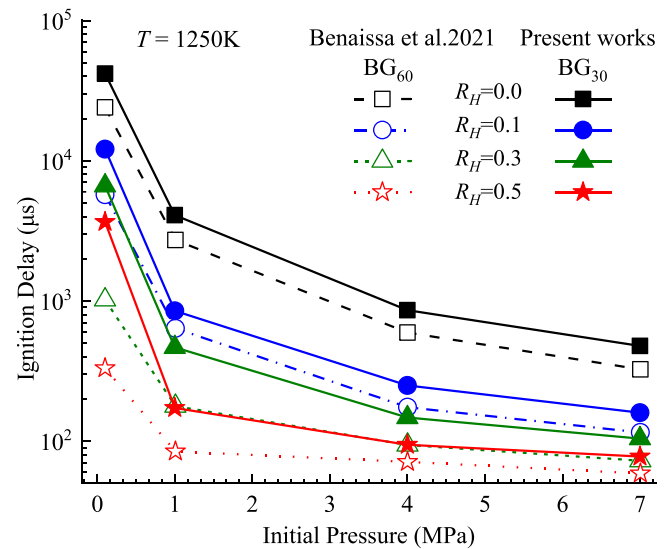
Fig. 8. Comparison of variation in mole fraction of H-radical (X_H) with normalized induction time (t/τ_i) with increase in hydrogen percentage in BG₃₀/hydrogen-air mixtures at initial $P = 4$ MPa and $T = 1000$ K & 1800 K.

the experimental results obtained from previous research studies show that the peak value of LBV shifts towards leaner mixtures. The simulation results obtained from the present study for low to medium calorific value fuel mixtures ($R_c=0.6$ to $R_c=0.8$) show the same trend. Thus, it can be concluded that the numerical model is capable of predicting combustion performance of fuel mixtures highly diluted with CO₂.

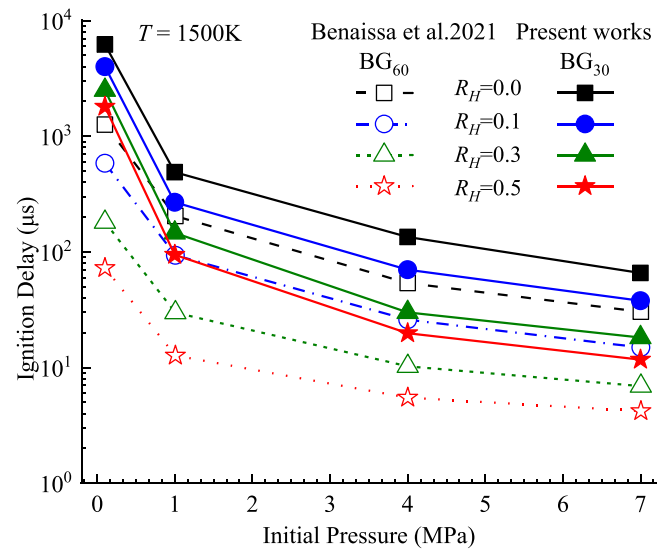
4. Results and discussion

4.1. Ignition delay

The ignition delay behavior of hydrogen enriched stoichiometric BG₃₀/hydrogen-air mixtures are investigated using the numerical model developed. The biogas, BG₃₀ comes under the category of low to medium calorific value LFG, which is the focus of this research study. Fig. 7a and b show the ignition delay results for four fuel mixtures with hydrogen enrichment of 0 %, 10 %, 30 % and 50 % at two pressures of 1 MPa and 4 MPa. The ignition delay results are plotted against $1000/T$ axis. Hence, 0.556, 0.625, 0.769 and 1.0 in x-axis represent initial



(a) T=1250K



(b) T=1500K

Fig. 9. Variation of ignition delay with an increase in hydrogen fraction added to BG₃₀/hydrogen-air mixtures at stoichiometric conditions against the initial pressure.

temperatures of 1800 K, 1600 K, 1300 K and 1000 K respectively.

As shown in the Fig. 7a and b, at two pressure conditions considered, BG₃₀ ($R_H=0.0$) shows almost a linear decrease in ignition delay time with increase in initial temperature from 1000 K to 1800 K. It is important to note here that the ignition delay time shows a logarithmic variation with the increase in initial temperature. With the hydrogen enrichment, the ignition delay time decreased as expected. However, in comparison to BG₆₀, the ignition delay time of BG₃₀ and Hydrogen enriched BG₃₀ decreased at significantly slower rates with the increase in initial temperature. For example, the data in Fig. 7(b) show that the ignition delay times of various mixtures of BG₃₀/hydrogen-air decreased approximately by 400 – 600 times when initial temperature increased from 1000 K to 1800 K. Under the same conditions, ignition delay time of BG₆₀ decreased by approximately 1000 – 5000 times (Benaissa et al., 2021). It shows that at higher initial temperatures, CO₂ dilution has a significant effect on the ignition delay.

According to the data in Fig. 7, at 1000 K initial temperature, a 10 %

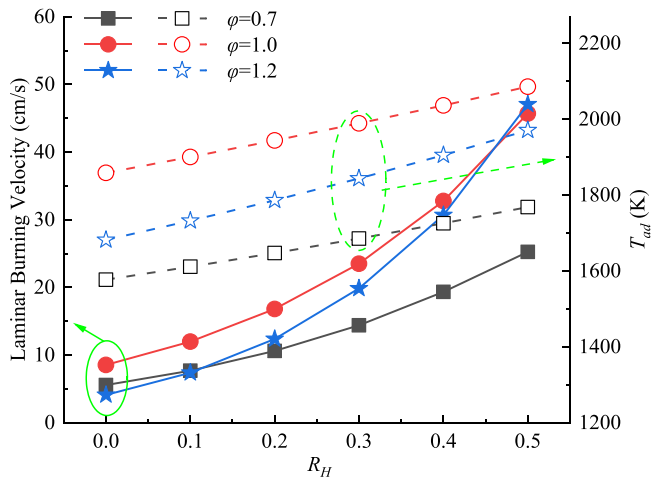


Fig. 10. Variation of laminar burning velocity, and adiabatic flame temperature (T_{ad}) of BG₃₀-air mixtures ($\phi = 0.7, 1.0$ and 1.2) against hydrogen addition at ambient conditions.

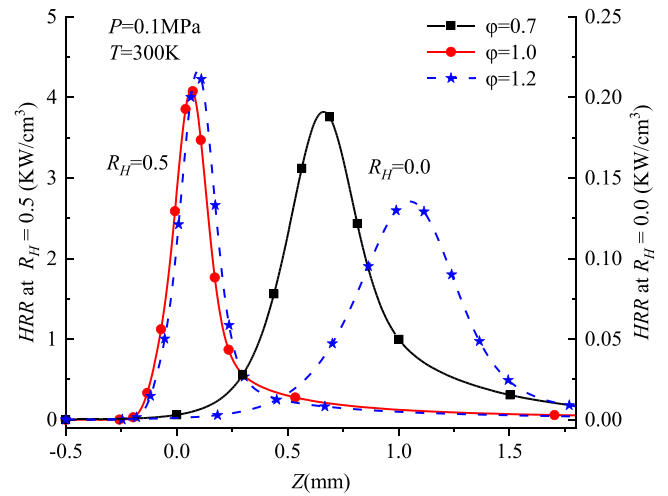
addition of hydrogen to BG₃₀ shows approximately 6 times reduction in ignition delay compared with pure BG₃₀. However, further increment of hydrogen fraction (up to $R_H = 0.5$) results in smaller improvement in ignition delay. The same phenomenon is reported for BG₆₀/hydrogen-air mixtures for the same range of hydrogen addition (Benaissa et al., 2021). According to Hu et al. (2019) H radical chain branching reactions (for example $H + CH_4 = CH_3 + H_2$ or $H + O_2 = O + OH$) are mainly responsible for ignition delay. The change in concentration of H radicals as given in Fig. 8 explains the pattern of variation in ignition delay of BG₃₀/hydrogen-air mixtures.

Fig. 8a and b show the comparison of variation in mole fraction of H-radical (X_H) with normalized induction time (t/τ_i) with increase in hydrogen percentage from $R_H = 0$ –0.5 in BG₃₀/hydrogen-air mixture at initial pressure, $P = 4$ MPa, and initial temperatures, $T = 1000$ K and $T = 1800$ K. The normalized induction time is obtained using the corresponding ignition delay time (τ_i) of a given fuel-oxidizer mixture. The data in Fig. 8(a) show that at initial temperature, $T = 1000$ K, the highest percentage increase (about 22 times from 0 % to 10 %, compared to about 1.3 times from 10 % to 30 % or from 30 % to 50 %) in H radical mole fraction is observed with the initial 10 % hydrogen addition in the BG₃₀-air mixture. Therefore, no significant improvement in ignition delay times were observed in the cases of $R_H = 0.3$ and 0.5 , compared to $R_H = 0.1$. For high-temperature ignition conditions (Fig. 8b), the percentage increase of H radical fraction is almost uniform. Therefore, ignition delay of BG₃₀ shows gradual improvement with the increase in hydrogen percentage in the mixture.

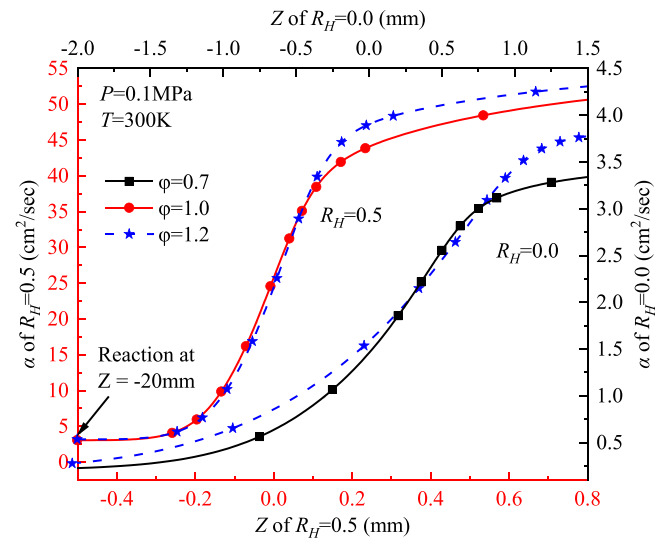
Fig. 9a and b show the ignition delay of BG₃₀/hydrogen-air (present work) and BG₆₀/hydrogen-air mixture (Benaissa et al., 2021) at stoichiometric condition with increasing initial pressure at two different initial temperatures ($T = 1250$ K and 1500 K). Both mixtures show substantial reduction of ignition delay in the pressure range of 0.1 MPa to 7.0 MPa. In comparison to BG₆₀/hydrogen-air mixtures, the rate of reduction of ignition delay is significantly high in BG₃₀/hydrogen-air mixtures within the same pressure range. According to the data in Fig. 9, for all fuel mixtures, the ignition delay time shows weak relationship with the initial pressure at high pressure conditions.

4.2. Laminar burning velocity and adiabatic flame temperature at ambient condition

Fig. 10 shows the variation of LBV and adiabatic flame temperatures (T_{ad}) for three different BG₃₀/hydrogen-air mixtures ($\phi = 0.7, 1.0$, and 1.2) at ambient conditions ($T = 300$ K and $P = 0.1$ MPa). In Fig. 10, dashed lines with open symbols represent T_{ad} and continuous lines with



(a) Heat release rate (HRR) KW/cm³



(b) Thermal diffusivity (α) cm²/sec

Fig. 11. Variation of Heat release rate profile (HRR) and Thermal diffusivity (α) along the flame thickness for two BG₃₀/hydrogen-air mixtures ($R_H = 0.0$ and $R_H = 0.5$) for $\phi = 0.7, 1.0$ and 1.2 for ambient conditions ($P = 0.1$ MPa & $T = 300$ K).

close symbols represents LBV. The data show that both the LBV and T_{ad} increased with increase in hydrogen fraction in BG₃₀/hydrogen-air mixtures. It is observed that the LBV of rich mixture ($\phi = 1.2$) and stoichiometric mixture ($\phi = 1.0$) become equal when $R_H \approx 0.45$. Similar behavior is also observed between rich mixture ($\phi = 1.2$) and lean mixture ($\phi = 0.7$) when $R_H \approx 0.1$. The reason for this behavior can be explained by studying the heat release rate (HRR) profile and the thermal diffusivity (α) along the flame thickness. Fig. 11a and b show the variation of HRR profile and α along the flame thickness for three equivalence ratios ($\phi = 0.7, 1.0$ and 1.2) at $R_H = 0.0$ and 0.5 for ambient conditions ($P = 0.1$ MPa and $T = 300$ K). The data show that both HRR and α increase for the rich case ($\phi = 1.2$) in comparison to stoichiometric case at $R_H = 0.5$ for ambient conditions. The comparison of average heat release rate (HRR_0) shows an increment of 4.7 % (about 80 W/cm³) with the increase in equivalence ratio from 1 to 1.2. The average thermal diffusivity (α_0) also increases by 18 % (about 0.5 cm²/s). In $R_H = 0$ mixture, HRR_0 increases while α_0 decreases for the rich case ($\phi = 1.2$) in comparison to lean case ($\phi = 0.7$). The HRR_0 decreased by 17.3 %

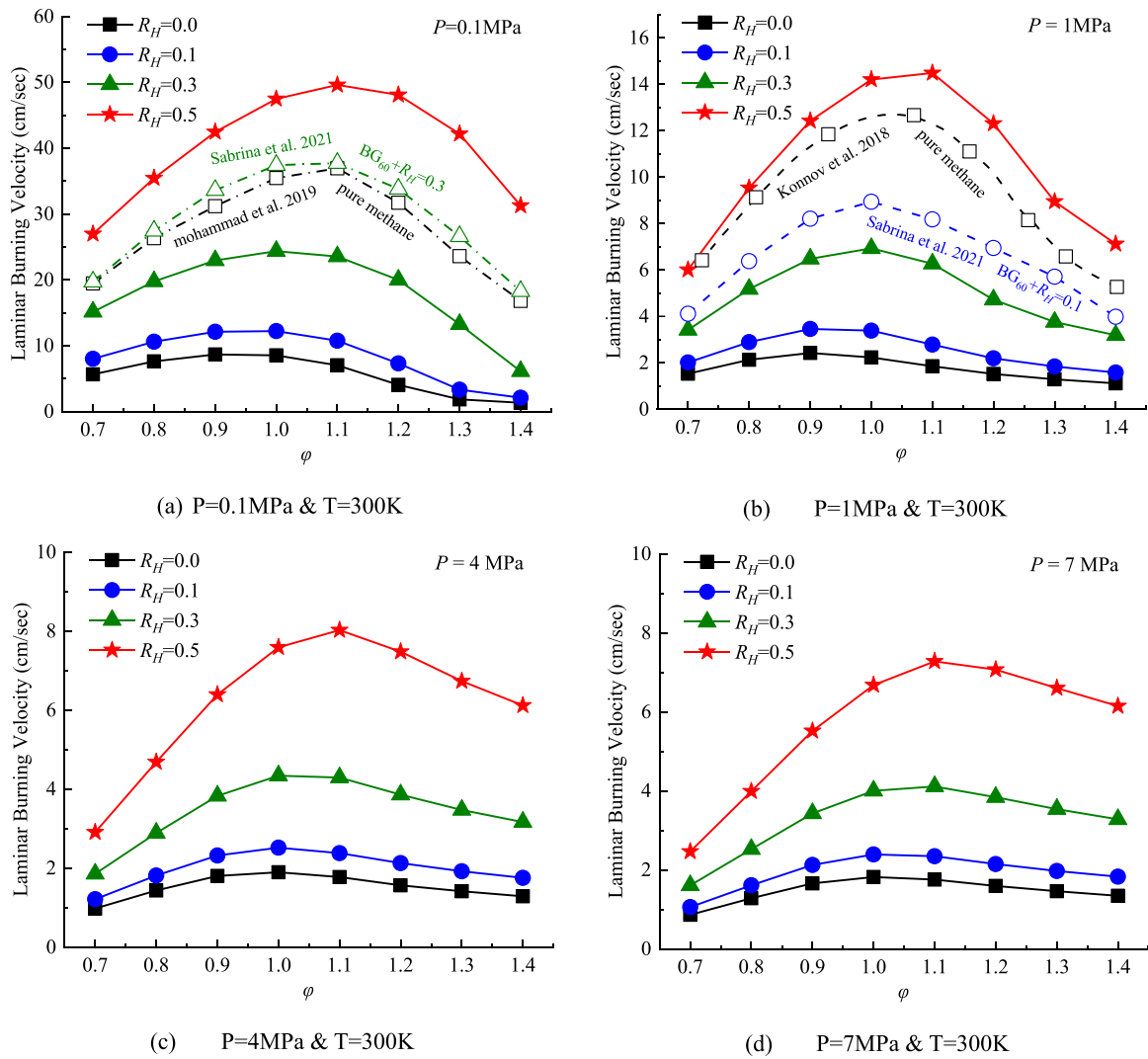


Fig. 12. Variation of the Laminar Burning Velocity of BG₃₀/hydrogen-air mixtures with increase in hydrogen fraction for various initial pressures (a) $P = 0.1$ MPa (b) $P = 1$ MPa (c) $P = 4$ MPa (d) $P = 7$ MPa & $T = 300$ K.

(about 10 W/cm^3) while α_0 increased by 6.1 % (about $0.17 \text{ cm}^2/\text{s}$). The net effect is that the square-root of the product of α_0 and HRR_0 ($\sqrt{\alpha_0 \text{HRR}_0}$) decreases leading to decrease in LBV (refer to Fig. 5 for more details).

The addition of hydrogen in the fuel mixture has a significant effect on the adiabatic flame temperature. The adiabatic flame temperature increased by 3–17 % as the R_H increased from 0 to 0.5. For the fuel mixtures with $\phi = 0.7, 1.0$, and 1.2 , the adiabatic flame temperature enhanced by 191 K, 227 K, and 289 K respectively with an increase in R_H from 0 to 0.5. It is therefore can be concluded that at ambient conditions, the hydrogen addition has a significant effect on adiabatic flame temperature, heat release rate and laminar burning velocity of the landfill gas composition which is categorized as low calorific value fuel where the diluent (CO_2) percentage is significantly high.

4.3. Laminar burning velocity at different initial pressures

Fig. 12(a) show the variation of LBV of different BG₃₀/hydrogen-air mixtures at ambient conditions ($P = 0.1$ MPa and $T = 300$ K) which is explained in more detail in the previous section. LBV variation of pure methane (Mohammad and Juhany, 2019) and BG₆₀/hydrogen-air ($R_H=0.3$) (Benaissa et al., 2021) at the same conditions are included for the comparison purpose. The LBV of BG₃₀/hydrogen-air ($R_H=0.3$) shows a significant improvement over BG₃₀/hydrogen-air ($R_H=0.1$) while

BG₃₀/hydrogen-air ($R_H=0.5$) shows much higher LBV than pure methane and BG₆₀/hydrogen-air ($R_H=0.3$) over the entire range of equivalence ratios considered. It shows that the low calorific value landfill gas could be enriched to obtain comparable or superior performance compared to pure methane by adding 30 – 50 % hydrogen.

Fig. 12b-d show the effect of changing initial pressure on the LBV of BG₃₀/hydrogen-air mixtures. The LBV decreased significantly with the increasing initial pressure. This effect can be visualized clearly by referring to Fig. 13 where variation of LBV with equivalence ratio at various initial pressures are presented for the 2 cases of $R_H=0$ and $R_H=0.4$ at $T=300$ K. In the literature, similar observations have been reported for BG₆₀ (Benaissa et al., 2021) and pure methane (Konnov et al., 2018; Mohammad and Juhany, 2019) for the range of conditions studied. In the present study, for 50 % hydrogen addition ($R_H=0.5$), maximum LBV decreases from 49.6 cm/s to less than 7.3 cm/s with the increase in initial pressure from 0.1 MPa to 7 MPa. It shows that hydrogen enrichment has an insignificant effect on the combustion performance of low calorific value landfill gases at high initial pressures. Therefore, it may be difficult or not practical to use hydrogen-enriched low calorific value landfill gases for applications where initial pressure is high.

In order to explore the observed pressure dependence of LBV further, a sensitivity analysis is performed to identify the significant elementary reactions affecting the LBV. As the LBV is proportional to square root of

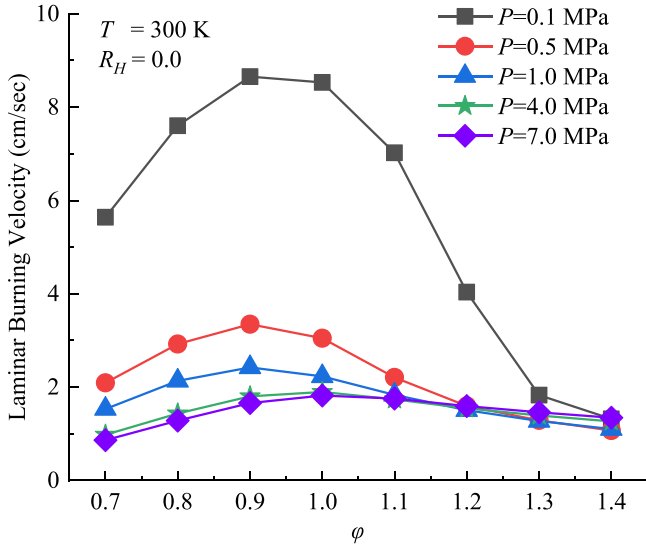
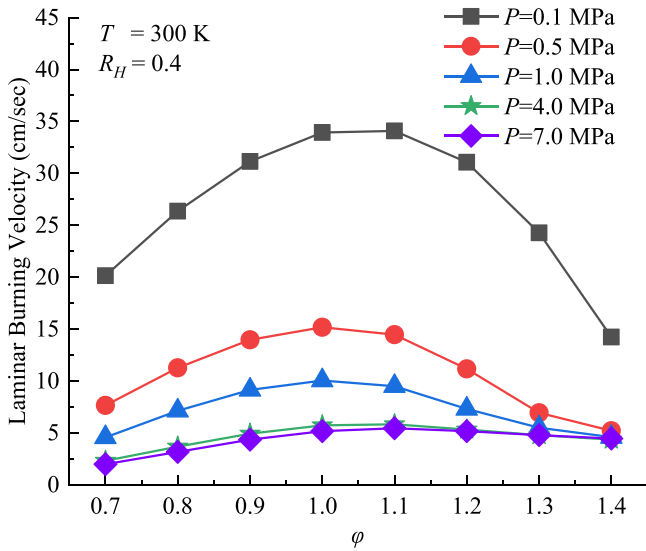
(a) $R_H=0.0$ & $T=300\text{K}$ (b) $R_H=0.4$ & $T=300\text{K}$

Fig. 13. The Laminar Burning Velocity of BG₃₀/hydrogen-air mixtures for various initial pressures (a) $R_H = 0.0$ (b) $R_H = 0.4$ at 300 K.

average thermal diffusivity and average reaction rate ($LBV \propto \sqrt{\alpha_0 \omega_0}$), the normalized sensitivity coefficient (S_{LBV}) used for the sensitivity analysis is given as follows:

$$S_{LBV} = \frac{\sqrt{\int R_{p,i} dx} - \sqrt{\int R_{0,i} dx}}{\sqrt{\int R_{0,i} dx}} \quad (5)$$

S_{LBV} is the normalized sensitivity coefficient of the i^{th} reaction, where $R = \alpha\omega$, the subscripts of $R_{0,i}$ and $R_{p,i}$ represent initial pressure of 0.1 MPa and elevated pressure for the i^{th} reaction, respectively. Fig. 14 a and b show the values of S_{LBV} for twelve important elementary reactions for BG₃₀-air ($R_H=0.0$) and BG₃₀/hydrogen-air ($R_H=0.4$) mixtures respectively. The numerical simulations were performed for four pressures (from 0.5 MPa to 7.0 MPa), at stoichiometric condition ($\phi=1.0$) and initial temperature of 300 K.

Generally, if the sensitivity coefficient of a main elementary reaction is positive, it indicates that the reaction accelerates the LBV; whereas the sensitivity coefficient of a main elementary reaction is negative, it

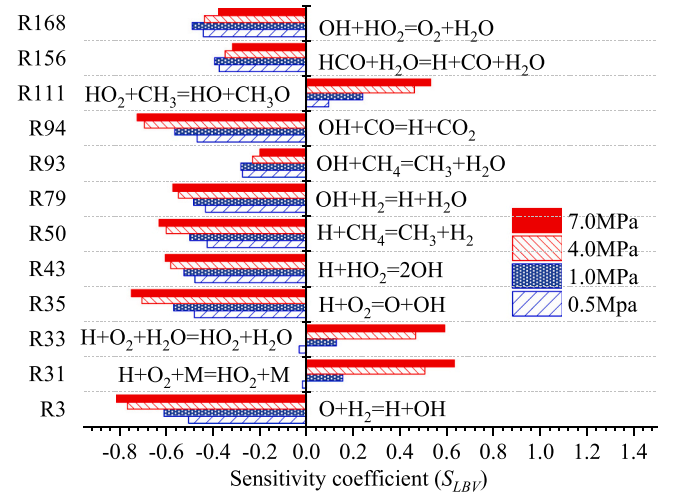
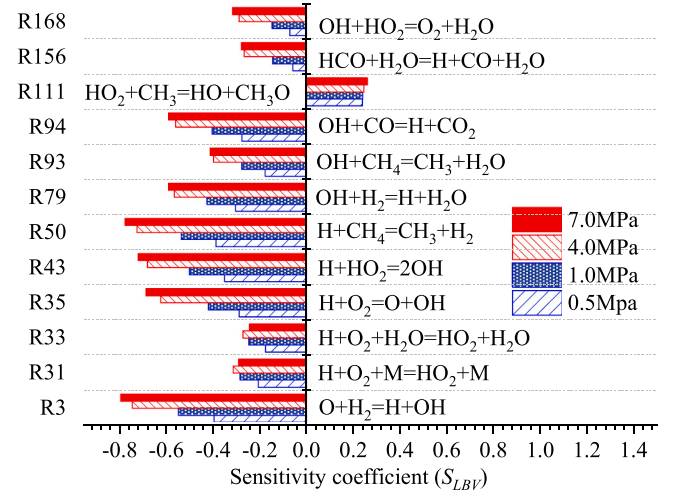
(a) $R_H=0.0$, $\phi=1.0$ and $T=300\text{K}$ (b) $R_H=0.4$, $\phi=1.0$ and $T=300\text{K}$

Fig. 14. Sensitivity coefficient of significant reaction pathways of BG₃₀-air ($R_H=0.0$) and BG₃₀/hydrogen-air ($R_H=0.4$) stoichiometric mixtures at different initial pressures (a) $R_H=0.0$, $\phi=1.0$ and $T=300\text{K}$ (b) $R_H=0.4$, $\phi=1.0$ and $T=300\text{K}$.

indicates that the reaction inhibits the LBV. Obviously, the data in Fig. 14 show that most of the twelve main reactions inhibit the LBV with increasing initial pressure in pure BG₃₀-air ($R_H=0.0$) and BG₃₀/hydrogen-air ($R_H=0.4$) mixtures. That is the reason why LBV decreases with increasing initial pressure. Only three reactions; R31: $\text{H} + \text{O}_2 + \text{M} = \text{HO}_2 + \text{M}$, R33: $\text{H} + \text{O}_2 + \text{H}_2\text{O} = \text{HO}_2 + \text{H}_2\text{O}$ and R111: $\text{HO}_2 + \text{CH}_3 = \text{HO} + \text{CH}_3\text{O}$ promote LBV in pure BG₃₀-air mixture (except at initial pressure of 0.5 MPa for R31 and R33) and just R111: $\text{H} + \text{O}_2 + \text{M} = \text{HO}_2 + \text{M}$ reaction promotes LBV in BG₃₀/hydrogen-air mixture ($R_H=0.4$).

It is interesting to note that the effect of reactions R31, R33 and R111 on LBV is more positive with increasing initial pressure, especially in pure BG₃₀-air mixture. And also, the effect of negative reactions R93: $\text{OH} + \text{CH}_4 = \text{CH}_3 + \text{H}_2\text{O}$, R156: $\text{HCO} + \text{H}_2\text{O} = \text{H} + \text{CO} + \text{H}_2\text{O}$ and R168: $\text{OH} + \text{HO}_2 = \text{O}_2 + \text{H}_2\text{O}$ on LBV is weakening with elevated pressure. That explains why LBV of high-pressure is greater than that of low-pressure at fuel-rich condition in Fig. 13(a).

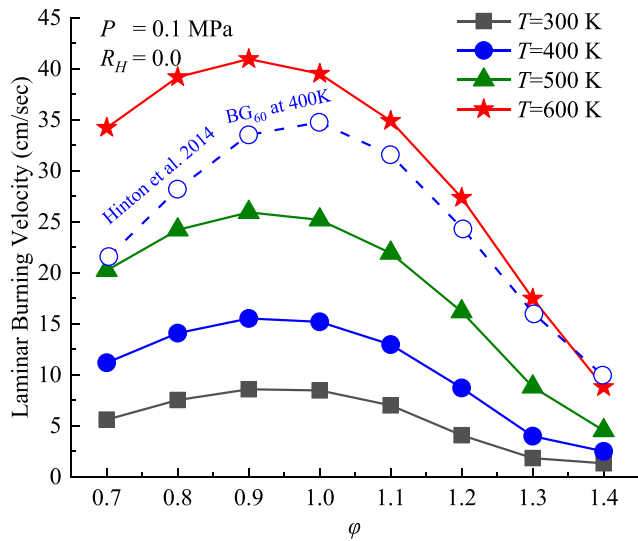
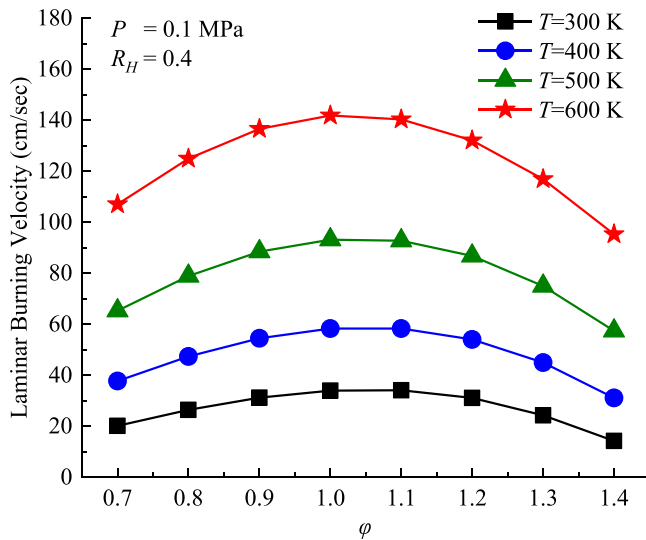
(a) BG_{30} -air ($R_H=0.0$)(b) BG_{30} /hydrogen-air ($R_H=0.4$)

Fig. 15. The laminar burning velocity of fuel mixtures at different initial temperatures and $P=0.1$ MPa (a) BG_{30} -air ($R_H=0.0$) (b) BG_{30} /hydrogen-air ($R_H=0.4$).

4.4. Laminar burning velocity at different initial temperatures

Fig. 15 shows how the LBV vary with the initial temperature for BG_{30} -air ($R_H=0.0$) and BG_{30} /hydrogen-air ($R_H=0.4$) over a range of equivalence ratios ($\phi=0.7$ – 1.4) and at $P=0.1$ MPa. The LBV of both fuel mixtures increased gradually with the increment of the initial temperature. When the initial temperature is 600 K, the LBV of BG_{30} -air ($R_H=0.0$) and BG_{30} /hydrogen-air ($R_H=0.4$) mixtures increased by around 7 and 5.3 times that of the corresponding fuel mixtures at ambient conditions. And also, the LBV of BG_{30} /hydrogen-air ($R_H=0.4$) shows higher sensitivity to initial temperature over the range of equivalence ratios considered. For example, LBV of stoichiometric BG_{30} /hydrogen-air ($R_H=0.4$) at initial temperature of 600 K is about 3.59 times more than that of BG_{30} -air ($R_H=0.0$) at initial temperature of 600 K and 16.7 times more than at initial temperature of 300 K. It can be observed that the initial temperature shows more linear influence on the LBV. As the initial temperature has a positive effect on the LBV, it can be

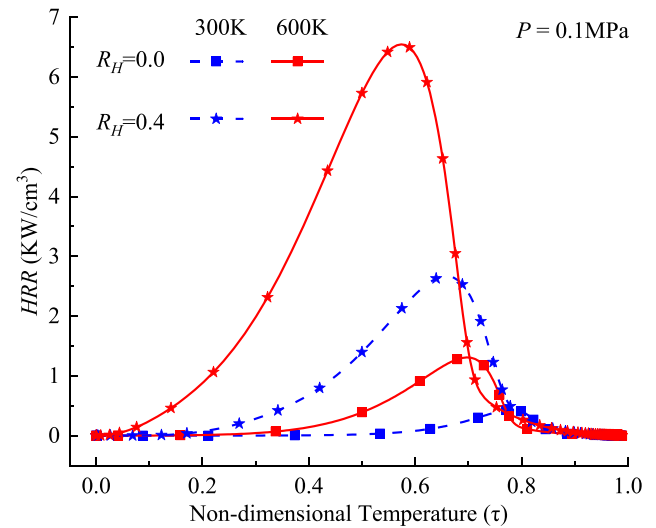
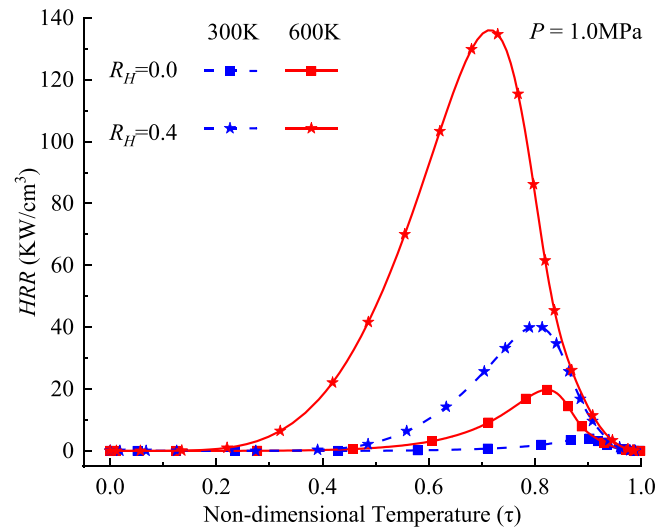
(a) $P = 0.1$ MPa(b) $P = 1$ MPa

Fig. 16. Heat release rates (HRR) of BG_{30} -air ($R_H=0.0$) and BG_{30} /hydrogen-air ($R_H=0.4$) mixtures against the non-dimensional temperature.

concluded that the negative effect of initial pressure on LBV may be effectively counteracted by operating the combustion systems at higher initial temperatures. Therefore, combustion systems designed to use hydrogen-enriched low calorific value landfill gases could be operated effectively at higher initial pressures by maintaining high initial temperature.

4.5. Study of effect of initial temperature on flame structure of BG_{30} /hydrogen-air mixtures

The change in the initial temperature conditions of fuel-air mixtures lead to change in resulting flame structures. It is due to the change of heat release rate and variations in major species profiles with increase in initial temperature. In previous research studies (Benaissa et al., 2021; Mohammad and Juhany, 2019), a non-dimensional temperature was used to investigate the effect of initial temperature on the flame structure. The non-dimensional temperature is defined as follows:

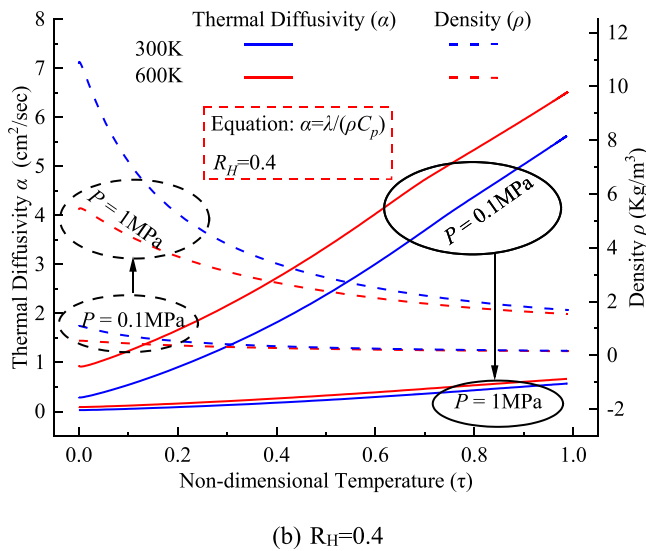
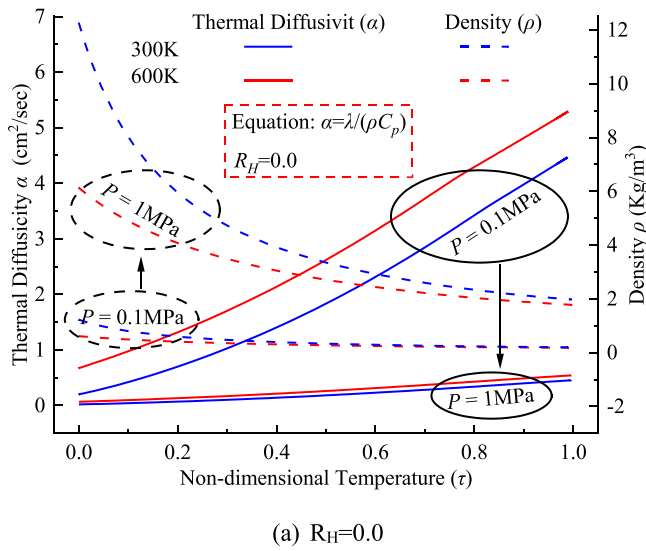


Fig. 17. Thermal diffusivity and density variation of BG₃₀-air ($R_H=0.0$) and BG₃₀/hydrogen-air ($R_H=0.4$) mixtures against the non-dimensional temperature at initial pressures of 0.1 MPa & 1 MPa, at initial temperatures of 300 K & 600 K.

$$\tau = \frac{T_i - T}{T_{ad} - T} \quad (6)$$

Where, T_i and T_{ad} are instantaneous temperature and adiabatic flame temperature respectively, and T is the initial temperature.

Fig. 16a and b show the variation of volumetric heat release rate (HRR) of stoichiometric mixtures of BG₃₀-air ($R_H=0.0$) and BG₃₀/hydrogen-air ($R_H=0.4$) against the non-dimensional temperature as defined in Eq. (6). It can be noted that the HRR increases with the increase in initial pressure and initial temperature. However, according to the Fig. 12 and Fig. 13, LBV of various fuel mixtures show a considerable reduction with the increase in initial pressure. This is due to the dominant effect of thermal conductivity of the dense fuel mixtures at higher pressures. The thermal conductivity of the fuel mixtures reduces as the pressure increases. It can be explained by comparing variation in thermal diffusivity and the density of BG₃₀-air and BG₃₀/hydrogen-air mixtures with the change in pressure as given in Fig. 17. As the pressure increased, there is a significant reduction of thermal diffusivity leading into reduction in thermal conductivity of the fuel mixtures. The overall effect is the reduction in LBV of fuel mixtures.

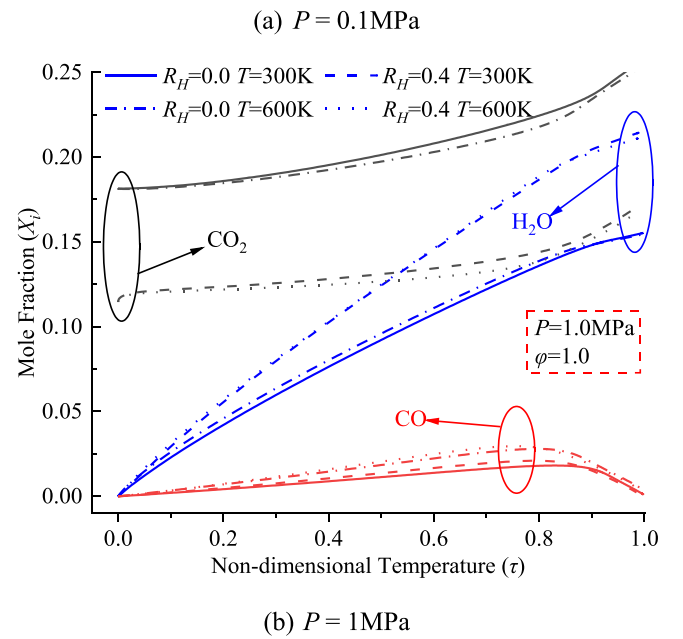
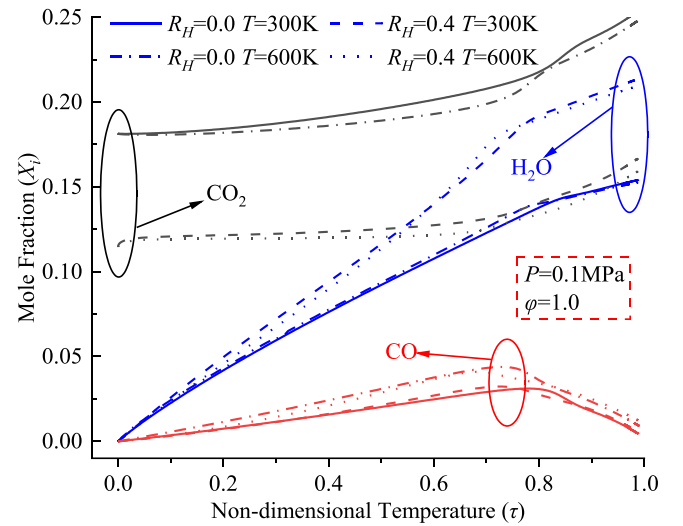
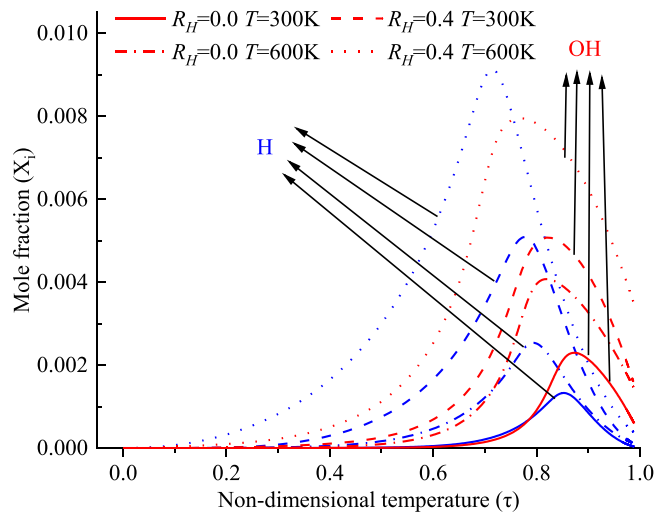
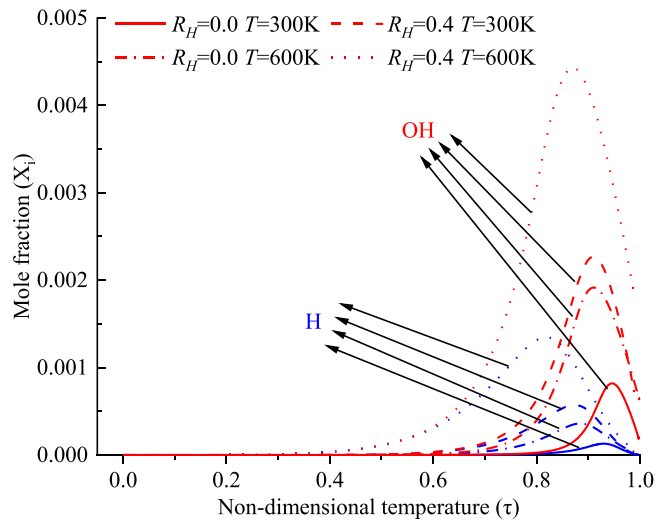


Fig. 18. Major species concentration against the non-dimensional temperature.

To study the effect of initial pressure and temperature on the combustion products of BG₃₀-air ($R_H=0.0$) and BG₃₀/hydrogen-air ($R_H=0.4$) mixtures, a combustion product analysis has been performed. Fig. 18a and b present the combustion product analysis (CO, CO₂ and H₂O) variation of the fuel mixtures considered at initial pressures of 0.1 MPa and 1 MPa and initial temperatures of 300 K and 600 K. The observed mole fractions of CO₂ is much higher than that of BG₆₀-air combustion (Benaissa et al., 2021), obviously because of the high percentage of CO₂ contained in BG₃₀. It is observed that the mole fraction of CO₂ decreased while that of H₂O increased as the initial temperature increased. This effect is common for both low and high initial pressures. It is due to the changes in reaction pathways with the change in initial temperature. The mole fractions of these species remain almost the same at elevated initial pressures. The variations of CO concentration under the conditions studied are insignificant.

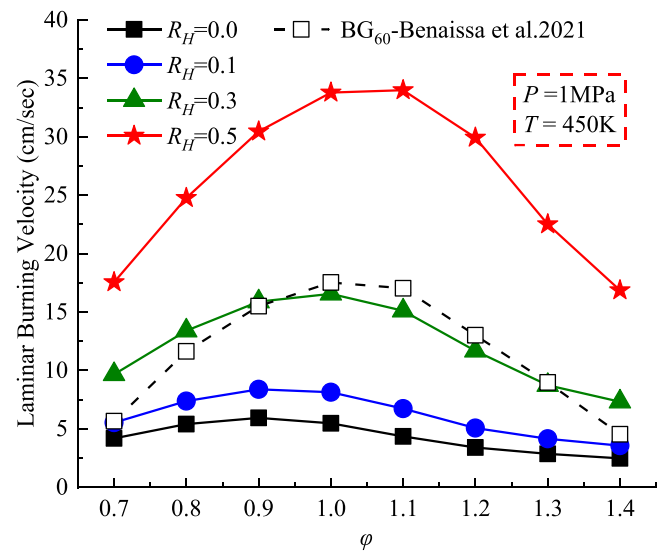
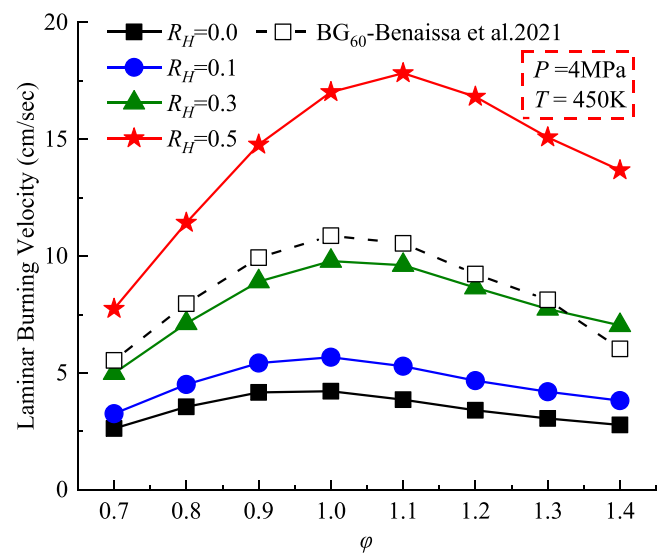
The minor species concentration presented in Fig. 19 show that the hydrogen fraction in the fuel mixture and the initial temperature have a significant influence on the production of minor species H and OH. The high concentration of radicals H and OH in the flame leads to high flame temperature and enhanced combustion. This explains the enhancement

(a) $P = 0.1 \text{ MPa}$ (b) $P = 1 \text{ MPa}$ **Fig. 19.** Minor species concentration against the non-dimensional temperature.

of LBV at higher initial temperatures and with the increase in hydrogen fraction in the fuel mixture. As the initial pressure increases there is a significant decrease in H and OH radical peaks. For BG₃₀/hydrogen-air ($R_H=0.4$), the observed decrease in H and OH radical peaks are about 6 and 1.5 times as the initial pressure increased from 0.1 MPa to 1 MPa. This observation is in line with the reduction of LBV with increased initial pressure and the importance of H and OH radical production/consumption reactions presented in sensitivity analysis in previous sections.

The data show that the LBV is positively correlated to initial temperature while it shows a negative correlation with initial pressure. As the combustion systems are operated at different initial temperatures and pressures, it is important to check the combined effect of higher initial temperature and pressure on LBV of various fuel mixtures. As a case example, LBV of BG₃₀/hydrogen-air mixtures at initial temperature of 450 K and two initial pressures of 1 MPa and 4 MPa were studied. These operating conditions were selected so that comparison can be made with corresponding results for BG₆₀ available in the literature.

As presented in Fig. 20(a) and (b), the maximum LBV of BG₃₀-air ($R_H=0.0$) and BG₃₀/hydrogen-air ($R_H=0.5$) at 450 K are 5 cm/s and 35 cm/s at 1 MPa. At 4 MPa, same values read as 3.5 cm/s and 17.5 cm/s

(a) $P=1 \text{ MPa} \text{ \& } T=450 \text{ K}$ (b) $P=4 \text{ MPa} \text{ \& } T=450 \text{ K}$ **Fig. 20.** Variation of the LBV of BG₃₀/hydrogen-air mixtures at elevated initial pressures (1 MPa and 4 MPa), and initial temperature, 450 K.

s respectively. Data show that hydrogen-enriched BG₃₀ ($R_H=0.3$) has comparable LBV to BG₆₀ at two conditions investigated. According to Fig. 21(a) and (b), the Heat Release Rate of the BG₆₀ is higher than that of BG₃₀/hydrogen ($R_H=0.3$) at the same operating conditions. However, the closeness of the data values indicates that the thermal energy produced by BG₃₀/hydrogen ($R_H=0.3$) fuel mixtures are sufficient to justify its use for heating applications. Therefore, it can be concluded that low calorific value LFG can probably be utilized as a viable alternative source of energy for industrial heating applications through hydrogen enrichment technique and reduce the potential environmental problems caused by current practices of managing low calorific value LFG. Combustion systems operated at low pressures and high initial temperatures are more suitable to use hydrogen-enriched low calorific value LFG. It should be verified further with experimental investigations.

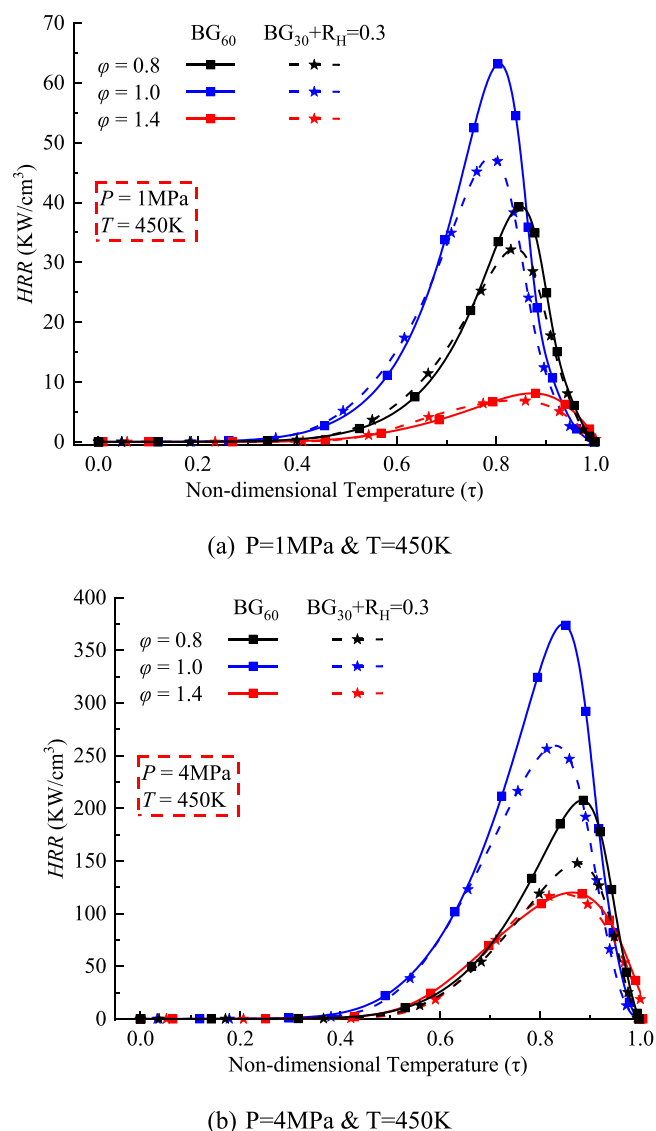


Fig. 21. Heat release rates (HRR) of BG₃₀/hydrogen-air ($R_H=0.3$) and BG₆₀-air against the non-dimensional temperature (a) at elevated initial pressures (1 MPa and 4 MPa), and initial temperature, 450 K.

5. Conclusions

In this study, combustion characteristics of hydrogen-enriched low calorific value LFG were investigated numerically to assess the potential of using the blended fuel for industrial thermal applications and hence enhance the utilization of low-quality LFG. Simulations were performed with ANSYS Chemkin-pro with GRI-Mech 3.0 mechanism. The validations performed for computational model show that the results are in good agreement with the experimental data available in the literature. The results show that the combustion characteristics of low calorific value LFG can be enhanced effectively through hydrogen enrichment technique and the use of suitably designed combustion systems. The major conclusions of the study are as follows:

1. The ignition delay time of BG₃₀/hydrogen-air mixtures decreased with the increase in initial temperature. However, in comparison to BG₆₀, the ignition delay time of BG₃₀ and hydrogen enriched BG₃₀ decreased at significantly slower rates with the increase in initial temperature. It shows that CO₂ dilution has a significant effect on the ignition delay. At 1000 K initial temperature, a 10 % addition of hydrogen to BG₃₀ shows approximately 6 times reduction in ignition

delay compared with pure BG₃₀. However, further increment of hydrogen fraction (up to $R_H=0.5$) results in smaller improvement in ignition delay. It is due to the fact that the highest percentage of H radical mole fraction occurs in the initial 10 % hydrogen addition and further increment of hydrogen creates small increment of H radical mole fraction.

2. The ignition delay time of BG₃₀/hydrogen-air mixtures also reduced with the increase of initial pressure. However, the ignition delay showed a weak relationship with the initial pressure in high pressure conditions.
3. The laminar burning velocity (LBV) and adiabatic flame temperature (T_{ad}) increased with increase in hydrogen fraction in BG₃₀/hydrogen-air mixtures at ambient conditions. It shows that the combustion performance of low calorific value LFG could be enhanced effectively through hydrogen enrichment technique at ambient conditions. However, the LBV decreased significantly with the increasing initial pressure. The hydrogen enrichment has an insignificant effect on the combustion performance of low calorific value LFG at high initial pressures. Therefore, it may be difficult or not practical to use hydrogen-enriched low calorific value LFG for applications with high initial pressure conditions.
4. The increment in initial temperature shows a positive effect on the LBV of BG₃₀/hydrogen-air mixtures. Therefore, it can be concluded that the negative effect of initial pressure on LBV may be effectively counteracted by operating the combustion systems at higher initial temperatures. Therefore, combustion systems designed to use hydrogen-enriched low calorific value LFG may be operated effectively at higher initial pressures by maintaining high initial temperatures.
5. The simulation results show that BG₃₀ enriched with 30 vol% hydrogen has comparable LBV to BG₆₀ for a range of operating conditions. The Heat Release Rate of the same fuel mixture is also sufficiently high. Therefore, the low calorific value LFG can probably be utilized as a viable alternative source of energy for industrial heating applications through hydrogen enrichment technique.

CRedit authorship contribution statement

Udaya Kahangamage: Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Kangdong Chen:** Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation. **Haisheng Zhen:** Writing – review & editing, Supervision, Conceptualization. **Chun-wah Leung:** Writing – review & editing, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The work described in this paper was fully supported by a grant from the Research Grants Council of the Hong Kong Special Administrative Region, China (UGC/FDS24/E11/22).

References

- Al-Hamamre, Z., Diezinger, S., Talukdar, P., Von Issendorff, F., Trimis, D., 2006. "Combustion of low calorific gases from landfills and waste pyrolysis using porous medium burner technology.", *Process Saf. Environ. Prot.* 84 (4), 297–308. <https://doi.org/10.1205/psep.05167>.

- An, Y., Tan, X., Gu, B., Zhu, K., Shi, L., Ding, Z., 2023. "An assessment of renewable energy development in Belt and Road Initiative countries: an entropy and TOPSIS approach," *Energy Rep.* 10, 3545–3560. <https://doi.org/10.1016/j.egy.2023.09.085>.
- Askari, M.H., Ashjaee, M., 2017. "Experimental measurement of laminar burning velocity and flammability limits of landfill gas at atmospheric and elevated pressures," *Energy Fuels* 31 (3), 3196–3205. <https://doi.org/10.1021/acs.energyfuels.6b02941>.
- Benaissa, S., Adouane, B., Ali, S.M., Mohammad, A., 2021. "Effect of hydrogen addition on the combustion characteristics of premixed biogas/hydrogen-air mixtures," *Int. J. Hydrog. Energy* 46 (35), 18661–18677. <https://doi.org/10.1016/j.ijhydene.2021.02.225>.
- Cardona, C.A., Amell, A.A., 2013. "Laminar burning velocity and interchangeability analysis of biogas/C₃H₈/H₂ with normal and oxygen-enriched air," *Int. J. Hydrog. Energy* 38 (19), 7994–8001. <https://doi.org/10.1016/j.ijhydene.2013.04.094>.
- Chandra, R., Vijay, V.K., Subbarao, P.M.V., Khura, T.K., 2011. "Performance evaluation of a constant speed IC engine on CNG, methane enriched biogas and biogas," *Appl. Energy* 88 (11), 3969–3977. <https://doi.org/10.1016/j.apenergy.2011.04.032>.
- Chen, Y., Kahangamage, U., Zhou, Q., Leung, C.W., 2021. "Can hydrogen enriched biogas be used as domestic fuel? - Part I – thermal characteristics of blended biogas/H₂ impinging flames," *HKIE Trans.* 28 (2), 60–67. <https://doi.org/10.33430/v28n2thie-2020-0040>.
- Chomiak, J., Longwell, J.P., Sarofim, A.F., 1989. "Combustion of low calorific value gases Problems and prospects," *Prog. Energy Combust. Sci.* 15 (2), 109–129. [https://doi.org/10.1016/0360-1285\(89\)90012-9](https://doi.org/10.1016/0360-1285(89)90012-9).
- Dai, W., Qin, C., Chen, Z., Tong, C., Liu, P., 2012. "Experimental studies of flame stability limits of biogas flame," *Energy Convers. Manag.* 63, 157–161. <https://doi.org/10.1016/j.enconman.2012.03.021>.
- M.G. David Brownell, Daniel Pawson, James Shaughnessy, "Control of landfill gas containing low concentrations of methane," Environment Agency, UK, Science Report – SC030305/SR2, 2009. [Online]. Available: (https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/291031/scho1008bouu-e-e.pdf).
- Duva, B.C., Chance, L.E., Toulson, E., 2020. "Effect of CO₂ dilution on the laminar burning velocities of premixed methane/air flames at elevated temperature," *J. Eng. Gas. Turbines Power* 142 (3). <https://doi.org/10.1115/1.4044641>.
- Hinton, N., Stone, R., 2014. "Laminar burning velocity measurements of methane and carbon dioxide mixtures (biogas) over wide ranging temperatures and pressures," *Fuel* 116, 743–750. <https://doi.org/10.1016/j.fuel.2013.08.069>.
- Hu, Y., Tan, J., Lv, L., Li, X., 2019. "Investigations on quantitative measurement of heat release rate using chemiluminescence in premixed methane-air flames," *Acta Astronaut.* 164, 277–286. <https://doi.org/10.1016/j.actaastro.2019.07.019>.
- Hu, Z., Zhang, X., 2019. "Experimental study on flame stability of biogas / hydrogen combustion," *Int. J. Hydrog. Energy* 44 (11), 5607–5614. <https://doi.org/10.1016/j.ijhydene.2018.08.011>.
- Kahangamage, U., Chen, Y., Leung, C.W., Ngai, T.Y., 2022. "Experimental Study of Lean-burning Limits of Hydrogen-enriched LPG Intended for Domestic Use," *J. Energy Power Technol.* 04 (02), 016. <https://doi.org/10.21926/jept.2202016>.
- Kahangamage, U., Chen, Y., Zhou, Q., Leung, C.W., 2021. "Can hydrogen enriched biogas be used as domestic fuel? - Part II: pollutants emission from combustion of biogas/H₂/air fuel mixture," *HKIE Trans.* 28 (2), 68–74. <https://doi.org/10.33430/v28n2thie-2020-0042>.
- Khan, A.R., Anbusaravan, S., Kalathi, L., Velamati, R., Prathap, C., 2017. "Investigation of dilution effect with N₂/CO₂ on laminar burning velocity of premixed methane/oxygen mixtures using freely expanding spherical flames," *Fuel* 196, 225–232. <https://doi.org/10.1016/j.fuel.2017.01.086>.
- Konnov, A.A., Mohammad, A., Kishore, V.R., Kim, N.I., Prathap, C., Kumar, S., 2018. "A comprehensive review of measurements and data analysis of laminar burning velocities for various fuel+air mixtures," *Prog. Energy Combust. Sci.* 68, 197–267. <https://doi.org/10.1016/j.pecs.2018.05.003>.
- Kumar, G., Kim, S.-H., Lay, C.-H., Ponnusamy, V.K., 2020. "Recent developments on alternative fuels, energy and environment for sustainability," *Bioresour. Technol.* 317, 124010. <https://doi.org/10.1016/j.biortech.2020.124010>.
- Kumar Yadav, V., Ray, A., Ravi, M.R., 2019a. "Experimental and computational investigation of the laminar burning velocity of hydrogen-enriched biogas," *Fuel* 235, 810–821. <https://doi.org/10.1016/j.fuel.2018.08.068>.
- Kumar Yadav, V., Ray, A., Ravi, M.R., 2019b. "Experimental and computational investigation of the laminar burning velocity of hydrogen-enriched biogas," *Fuel* 235, 810–821. <https://doi.org/10.1016/j.fuel.2018.08.068>.
- Lee, S., Kwon, O.C., 2011. "Effects of ammonia substitution on extinction limits and structure of counterflow nonpremixed hydrogen/air flames," *Int. J. Hydrog. Energy* 36 (16), 10117–10128.
- Li, J., et al., 2017. "Combustion and heat release characteristics of biogas under hydrogen- and oxygen-enriched condition," *Energy* 10 (8), 1200. <https://doi.org/10.3390/en10081200>.
- Liang, M., He, Y., Liao, S., Jian, X., Shao, Y., 2021. "Coupling effects of hydrogen transport and reactivity on NO production in laminar nonpremixed methane/air flames," *Fuel* 288, 119725. <https://doi.org/10.1016/j.fuel.2020.119725>.
- Liu, G., Zhou, J., Wang, Z., Yang, W., Liu, J., Cen, K., 2020. "Adiabatic laminar burning velocities of C₃H₈-O₂-CO₂ and C₃H₈-O₂-N₂ mixtures at ambient conditions-PART II: mechanistic interpretation," *Fuel* 276. <https://doi.org/10.1016/j.fuel.2020.117946>.
- Mitu, M., Giurcan, V., Razus, D., Oancea, D., 2017. "Inert gas influence on the laminar burning velocity of methane-air mixtures," *J. Hazard Mater.* 321, 440–448. <https://doi.org/10.1016/j.jhazmat.2016.09.033>.
- Mohammad, A., Juhany, K.A., 2019. "Laminar burning velocity and flame structure of DME/methane+air mixtures at elevated temperatures," *Fuel* 245, 105–114. <https://doi.org/10.1016/j.fuel.2019.02.085>.
- Qin, W., Egolfopoulos, F.N., Tsotsis, T.T., 2001. "Fundamental and environmental aspects of landfill gas utilization for power generation," *Chem. Eng. J.* 82 (1), 157–172. [https://doi.org/10.1016/S1385-8947\(00\)00366-1](https://doi.org/10.1016/S1385-8947(00)00366-1).
- Rajesh, N., Prathap, C., 2023. "Effect of hydrogen addition on the laminar burning velocity and the flame stability of n-dodecane reacting with air at elevated pressures," *Int. J. Hydrog. Energy*. <https://doi.org/10.1016/j.ijhydene.2023.07.077>.
- Stančin, H., Mikulić, H., Wang, X., Duić, N., 2020. "A review on alternative fuels in future energy system," *Renew. Sustain. Energy Rev.* 128, 109927. <https://doi.org/10.1016/j.rser.2020.109927>.
- Striugas, N., Paulauskas, R., Skvorčinskienė, R., Lisauskas, A., 2020a. "Investigation of waste biogas flame stability under oxygen or hydrogen-enriched conditions," *Energy* 13 (18), 4760. <https://doi.org/10.3390/en13184760>.
- Striugas, N., Zakarauskas, K., Paulauskas, R., Skvorčinskienė, R., 2020b. "Chemiluminescence-based characterization of tail biogas combustion stability under syngas and oxygen-enriched conditions," *Exp. Therm. Fluid Sci.* 116, 110133. <https://doi.org/10.1016/j.expthermflusci.2020.110133>.
- U. S. E. P. A. (EPA). "Basic Information about Landfill Gas." (<https://www.epa.gov/lmop/basic-information-about-landfill-gas>) (accessed 19/01/2024).
- Uchman, W., Werle, S., 2016. "The use of low-calorific value gases in environmental protection engineering," *Archit., Civ. Eng., Environ.* 9 (1), 127–132. <https://doi.org/10.21307/acee-2016-014>.
- Wang, Z., Wang, S., Whiddon, R., Han, X., He, Y., Cen, K., 2018. "Effect of hydrogen addition on laminar burning velocity of CH₄/DME mixtures by heat flux method and kinetic modeling," *Fuel* 232, 729–742. <https://doi.org/10.1016/j.fuel.2018.05.146>.
- Wei, Z.L., Leung, C.W., Cheung, C.S., Huang, Z.H., 2016. "Effects of H₂ and CO₂ addition on the heat transfer characteristics of laminar premixed biogas–hydrogen Bunsen flame," *Int. J. Heat. Mass Transf.* 98, 359–366. <https://doi.org/10.1016/j.ijheatmasstransfer.2016.02.064>.
- Wei, Z., Zhen, H., Fu, J., Leung, C., Cheung, C., Huang, Z., 2019. "Experimental and numerical study on the laminar burning velocity of hydrogen enriched biogas mixture," *Int. J. Hydrog. Energy* 44 (39), 22240–22249. <https://doi.org/10.1016/j.ijhydene.2019.06.097>.
- Wei, Z., Zhen, H., Leung, C., Cheung, C., Huang, Z., 2020. "Effects of H₂ addition on the formation and emissions of CO/NO₂/NO_x in the laminar premixed biogas-hydrogen flame undergoing the flame-wall interaction," *Fuel* vol. 259, 116257. <https://doi.org/10.1016/j.fuel.2019.116257>.
- Yuyin, Z., Jianghong, W.U., Ishizuka, S., 2009. "Hydrogen addition effect on laminar burning velocity, flame temperature and flame stability of a planar and a curved CH₄-H₂-air premixed flame," *Int. J. Hydrog. Energy* 34 (1), 519–527. <https://doi.org/10.1016/j.ijhydene.2008.10.065>.
- Zeng, W., Ma, H., Liang, Y., Hu, E., 2015. "Experimental and modeling study on effects of N₂ and CO₂ on ignition characteristics of methane/air mixture," *J. Adv. Res.* 6 (2), 189–201. <https://doi.org/10.1016/j.jare.2014.01.003>.
- Zhang, C., Hu, G., Liao, S., Cheng, Q., Xiang, C., Yuan, C., 2016. "Comparative study on the effects of nitrogen and carbon dioxide on methane/air flames," *Energy* 106, 431–442. <https://doi.org/10.1016/j.energy.2016.03.087>.
- Zhen, H.S., Zhang, L., Wei, Z.L., Chen, Z.B., Huang, Z.H., 2019. "A numerical study of the heat transfer of an impinging round-jet methane Bunsen flame," *Fuel* 251, 730–738. <https://doi.org/10.1016/j.fuel.2019.04.077>.
- Zhou, J., Liu, G., Wang, Z., Yang, W., Liu, J., Cen, K., 2020. "Adiabatic laminar burning velocities of C₃H₈-O₂-CO₂ and C₃H₈-O₂-N₂ mixtures at ambient conditions-PART I: experimental and numerical study," *Fuel* 263. <https://doi.org/10.1016/j.fuel.2019.116533>.